

USER MANUAL

FOR THE

PREPARATION OF

WATER SECURITY PLAN

DRAFT

Government of India
Ministry of Jal Shakti,
Department of Water Resources,
River Development and Ganga Rejuvenation

CONTENT

A. PRE PLANNING ACTIVITIES BY DPMU/SPMU BEFORE INITIATING THE WATER SECURITY PLAN PREPARATION 1

B. PREPARATION OF WATER SECURITY PLAN 2

I. FLOW CHART FOR PREPARATION OF WATER SECURITY PLAN 2

II. SUB - MODULES OF WATER SECURITY PLAN 3

1.GRAM PANCHAYAT (BASIC HYDROLOGY INFO) SUB MODULE 5

2.WATER AVAILABILITY SUB MODULE 7

3.WATER UTILIZATION SUB MODULE 13

4.WATER BALANCE SUB MODULE 23

5.WATER BUDGET SUB MODULE 25

6.DEMAND SIDE SUB MODULE 28

7.SUPPLY SIDE SUB MODULE 37

8.PGWM COMMITTEE SUB MODULE 41

9.WSP REMARKS 43

10.FINAL WATER SECURITY PLAN REPORT 44

DRAFT

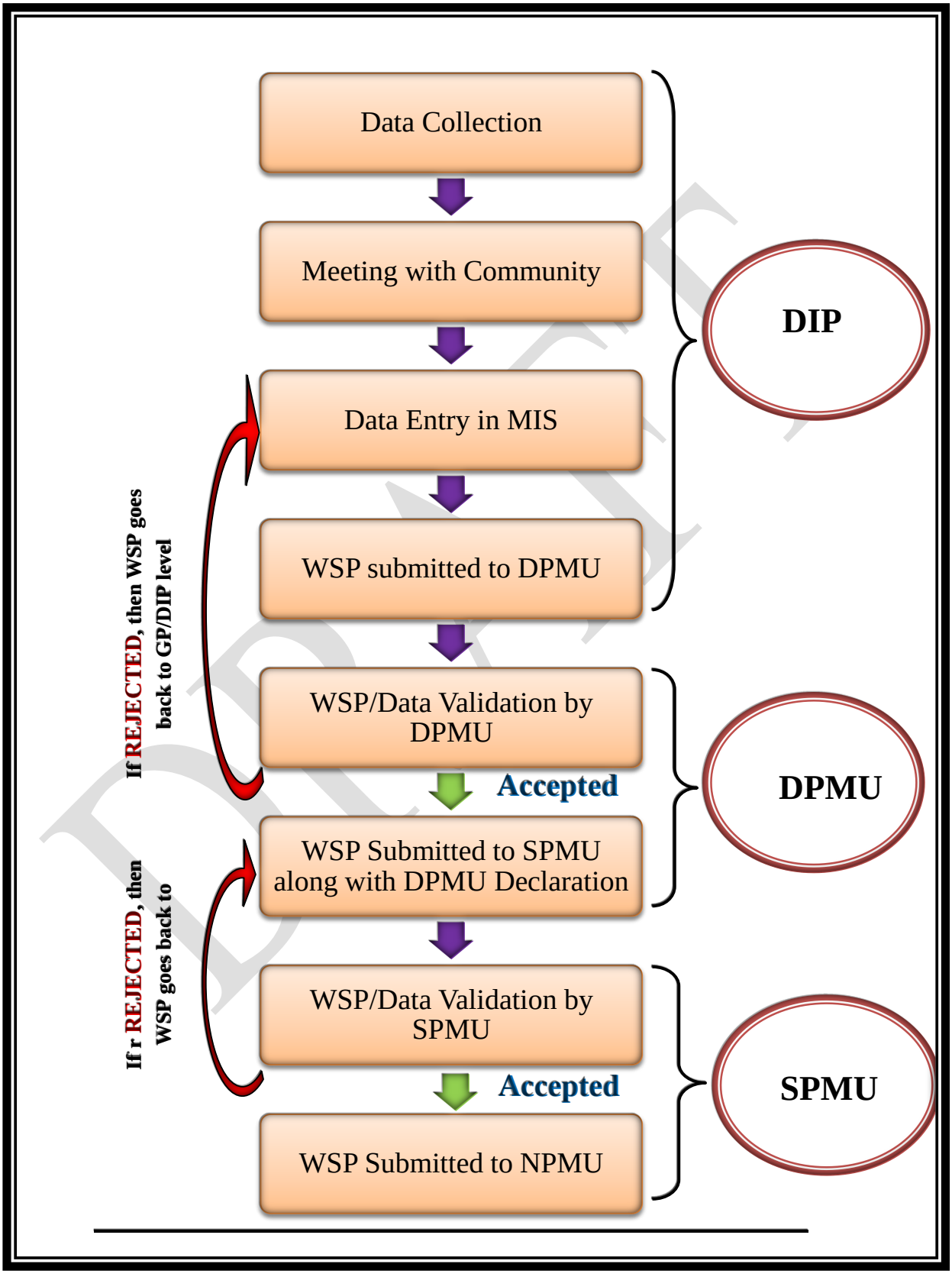
A. PRE PLANNING ACTIVITIES BY DPMU/SPMU BEFORE INITIATING THE WATER SECURITY PLAN PREPARATION

1) Guideline version 1.1 for WSP to be translated in local language.
2) All 3 mandatory maps to be prepared at the DPMU/SPMU level.
3) DPMU Nodal officer / DM shall write a letter to the concerned Block level officer with a copy to Sarpanch and Panchayat Secretary regarding WSP preparation and collection of data.
4) A letter to be issued from DPMU Nodal (DM/DC) to Sarpanch/ President of VWSC/ PGWM regarding updation of WSP for the year 2023.
5) DPMU to chalk out a plan for training of DIPs for preparation of WSP.
6) Letter from DPMU Nodal to all Line Departments for sharing the data with DIPs.
7) DPMU to issue a letter of identification to all the DIP members.

DRAFT

B. PREPARATION OF WATER SECURITY PLAN

I.FLOW CHART FOR PREPARATION OF WATER SECURITY PLAN

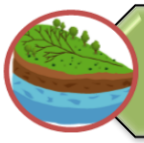


II. SUB - MODULES OF WATER SECURITY PLAN

DRAFT



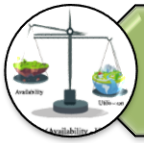
Gram Panchayat
(Basic Hydrology Info)



Water Availability



Water Utilization



Water Balance



Water Budget



Demand Side



Supply Side



PGWM Committee



WSP Remarks



Final Water Security Plan

DRAFT

1. GRAM PANCHAYAT (BASIC HYDROLOGY INFO) SUB MODULE

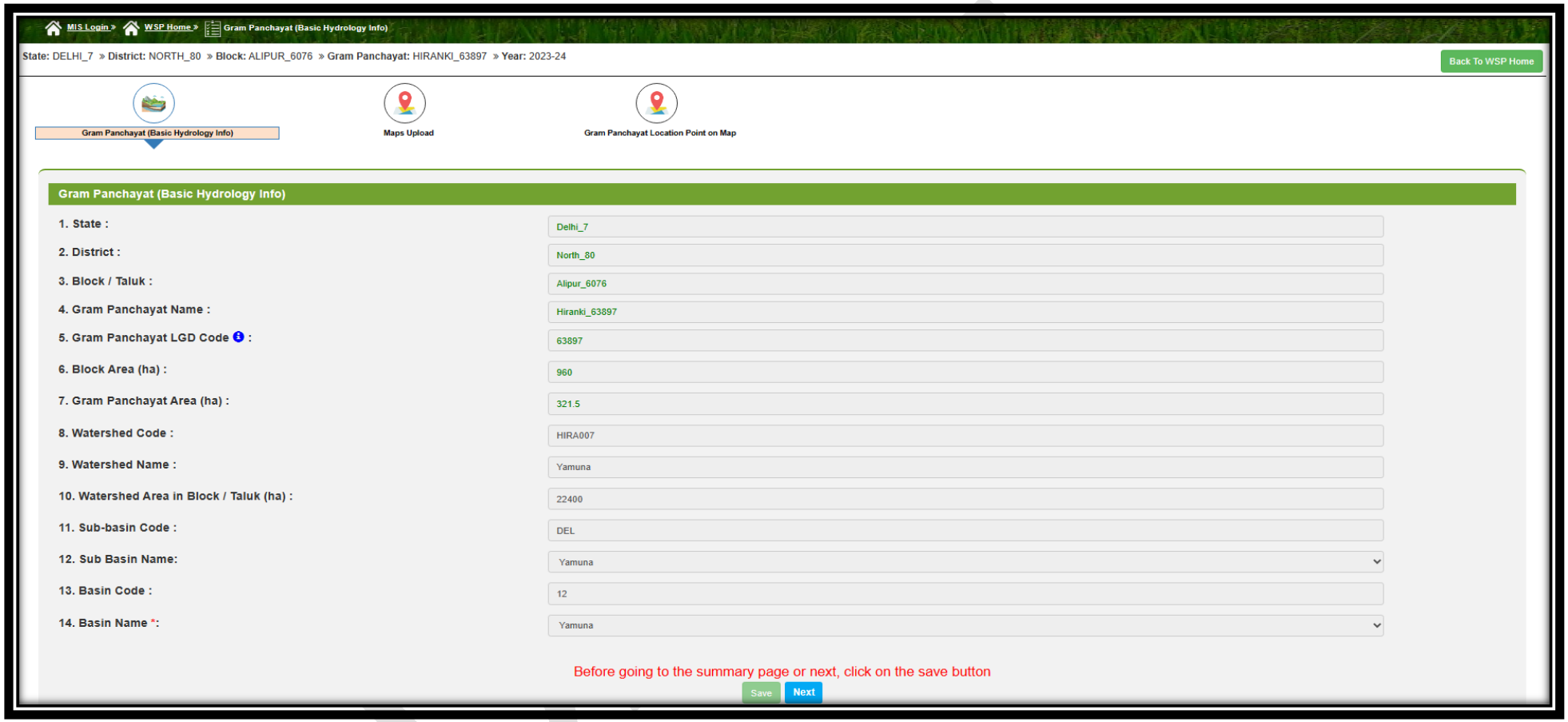


Fig: 1.1

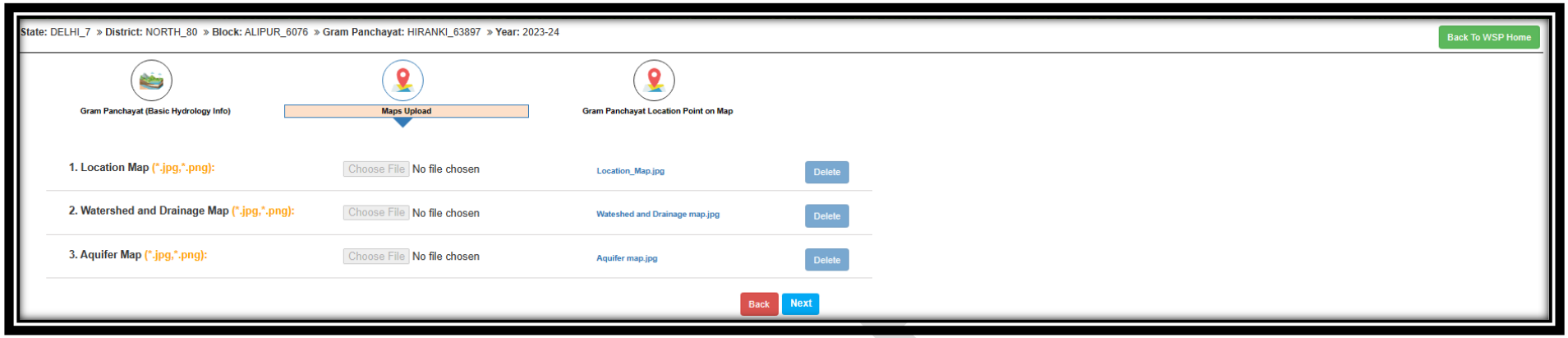


Fig:1.2

TABLE:1

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom
1	1-7	All details are captured from GPDCI module / Backend Data Table	GPDCI/ Backend Data Table	Auto Fetched
2	8-14	Hydrological details to be entered	Watershed Map/Atlas of CGWB	DIP/DPMU
3	Maps Upload	Location Map, Watershed and Drainage Map, Aquifer Map	Watershed and Drainage Map & Aquifer Map - CGWB	DIP/DPMU/SPMU

DRAFT

2. WATER AVAILABILITY SUB MODULE

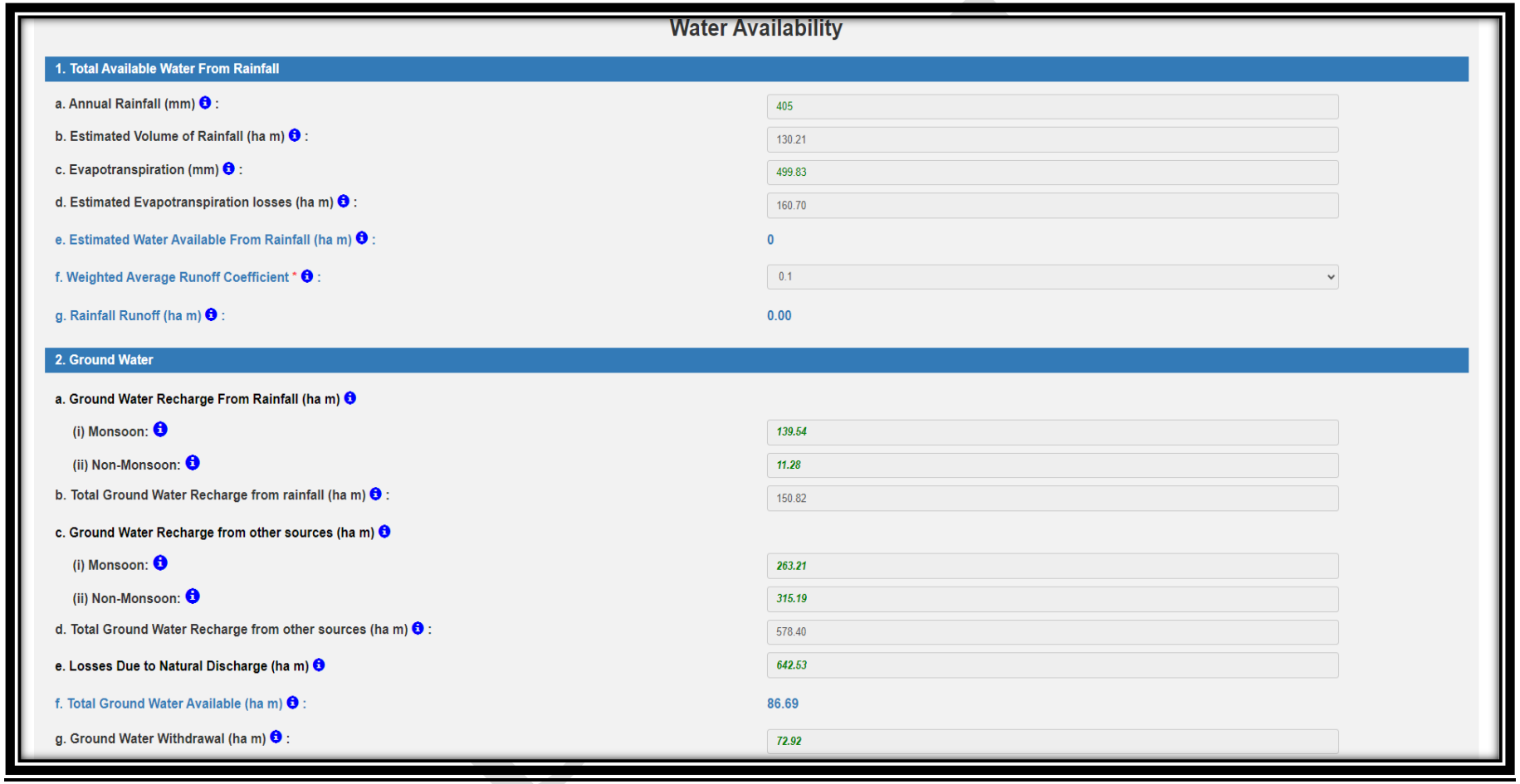


Fig: 2.1





3. Surface Water

a. Surface Water available in storages

(i) Number of Surface Water Bodies and Storage Capacity (ha m):

Note:- Enter Sum of storage capacity, if same type of structure are more than one.

Add

Type of Structure	No. of Structures	Storage Capacity (ha m)	
Ponds	5	10	Remove
Tanks	2	5	Remove

(ii) Total Storage Capacity of the surface water bodies (ha m) :	15
(iii) Losses if Any (%) :	10
(iv) Surface Water Available in Storages (ha m) :	13.50
b. Transported Water from outside GP area (ha m) :	5
c. Total Surface Water Available (ha m) :	18.50

4. Total Water Available

Total Water Available (ha m) :	105.19

Before going to the summary page or next, click on the save button

Back Save Summary

Fig:2.2

TABLE:2

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	1 (a)	Annual Rainfall Data (mm) Period: January- December , 2022	IMD (India WRIS)	Auto fetched from backend however manual entry is also allowed provided that user has an authentic source	
2	1 (b)	Estimated Volume of Rainfall (ha m)		Auto Calculated	GP Area (ha) * Annual Rainfall (mm) * 0.001
3	1 (c)	Evapotranspiration Data (mm) Period: June-October, 2022	India WRIS	Auto fetched from backend however manual entry is also allowed provided that user has an authentic source	
4	1 (d)	Estimated Evapotranspiration losses (ha m)		Auto Calculated	GP Area (ha) * Evapotranspiration (mm) * 0.001
5	1 (e)	Estimated Water Available from Rainfall (ha m)		Auto calculated	Estimated Volume of Rainfall (ha m)-Estimated Evapotranspiration losses (ha m)
6	1 (f)	Weighted Average Runoff Coefficient; Select from Dropdown	Table 4.4, Page 25 of Manual on Artificial Recharge of GW,CGWB, 2007	DIP/DPMU	

DRAFT

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
7	1 (g)	Rainfall Runoff (ha m) Note: The Volume of Supply Side Interventions to be restricted up to less than 50% of rainfall runoff.		Auto Calculated	Estimated Water Available from Rainfall (ha m) * Weighted Average Runoff Coefficient
8	2 (a) (i)	Ground Water Recharge from Rainfall (ha m) (Monsoon) ; Apportioned to GP Area as per GW assessment Report-2022, CGWB	Block wise GWRA-2022	Auto filled entry however manual entry is also allowed provided that user has an authentic source	Block wise recharge from Rainfall- Monsoon* GP Area/Block Area
9	2 (a) (ii)	Ground Water Recharge from Rainfall (ha m) (Non-Monsoon); Apportioned to GP Area as per GW assessment Report-2022, CGWB	Block wise GWRA-2022	Auto filled entry however manual entry is also allowed provided that user has an authentic source	Block wise recharge from Rainfall- Non Monsoon* GP Area/Block Area
10	2 (b)	Total Ground Water Recharge from Rainfall (ha m)		Auto Calculated	Monsoon+ Non Monsoon
11	2 (c) (i)	Ground Water Recharge from Other Sources (ha m) (Monsoon) ; Apportioned to GP Area as per GW assessment Report-2022,CGWB	Block wise GWRA-2022	Auto filled entry however manual entry is also allowed provided that user has an authentic source	Block wise recharge from other sources - Monsoon* GP Area/Block Area
12	2 (c) (ii)	Ground Water Recharge from Other Sources (ha m) (Non-Monsoon); Apportioned to GP Area as per GW assessment Report-2022,CGWB	Block wise GWRA-2022	Auto filled entry however manual entry is also allowed provided that user has an authentic source	Block wise recharge from other sources- Non Monsoon* GP Area/Block Area

DRAFT

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
13	2 (d)	Total Ground Water Recharge from other sources (ha m)		Auto Calculated	Monsoon (ha m)+ Non Monsoon (ha m)
14	2 (e)	Losses due to Natural Discharge (ha m); Apportioned to GP Area as per GW assessment Report-2022	Block wise GWRA-2022	Auto filled entry manual entry is also allowed provided that user has an authentic source	Block wise Losses due to Natural Discharge * GP Area/ Block Area
15	2 (f)	Total Ground Water Available (ha m)		Auto Calculated	Total Ground Water Recharge from Rainfall (ha m) + Total Ground Water Recharge from other sources (ha m) - Losses due to natural discharge (ha m)
16	2 (g)	Ground Water withdrawal (ha m); Apportioned to GP Area as per GW assessment Report-2022	Block wise GWRA-2022	Auto filled entry however manual entry is also allowed provided that user has an authentic source	Block wise Ground Water Withdrawal * GP Area/ Block Area
17	3(a) (i)	Details of no. of surface water bodies and storage capacity to be entered	Line Department	DIP/DPMU	
18	3(a) (ii)	Total Storage Capacity of the Surface Water Bodies (ha m)		Auto Calculated	Sum of capacities of individual structures excluding Rain Water Harvesting Structures
19	3(a) (iii)	Losses if Any (%); Select from dropdown	Line Department	DIP/DPMU	
20	3(a) (iv)	Surface Water Available in Storages (ha m)		Auto Calculated	Total Storage Capacity of Surface Water bodies (ha m) – [Total Storage Capacity of Surface Water bodies (ha m) * (Losses if any(%)/100)]

DRAFT

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
21	3 (b)	Transported Water from outside GP area (ha m); Transported water including water available from canal/ river lift. Consider transported water which used for irrigation or domestic purposes only.	Line Department; Transported water at GP level to be worked out by apportioning of total inflow through canal/lift at Head Reach	DIP	
22	3 (c)	Total Surface Water Available (ha m)		Auto Calculated	Surface Water Available in Storages (ha m) + Transported Water from outside GP area (ha m) + Water Available from Canal/ River lift (ha m)
23	4	Total Water Available (ha m)		Auto Calculated	Total Ground Water Available (ha m) + Total Surface Water Available (ha m)

DRAFT

3. WATER UTILIZATION SUB MODULE

3.1: Drinking/Domestic Water Utilization

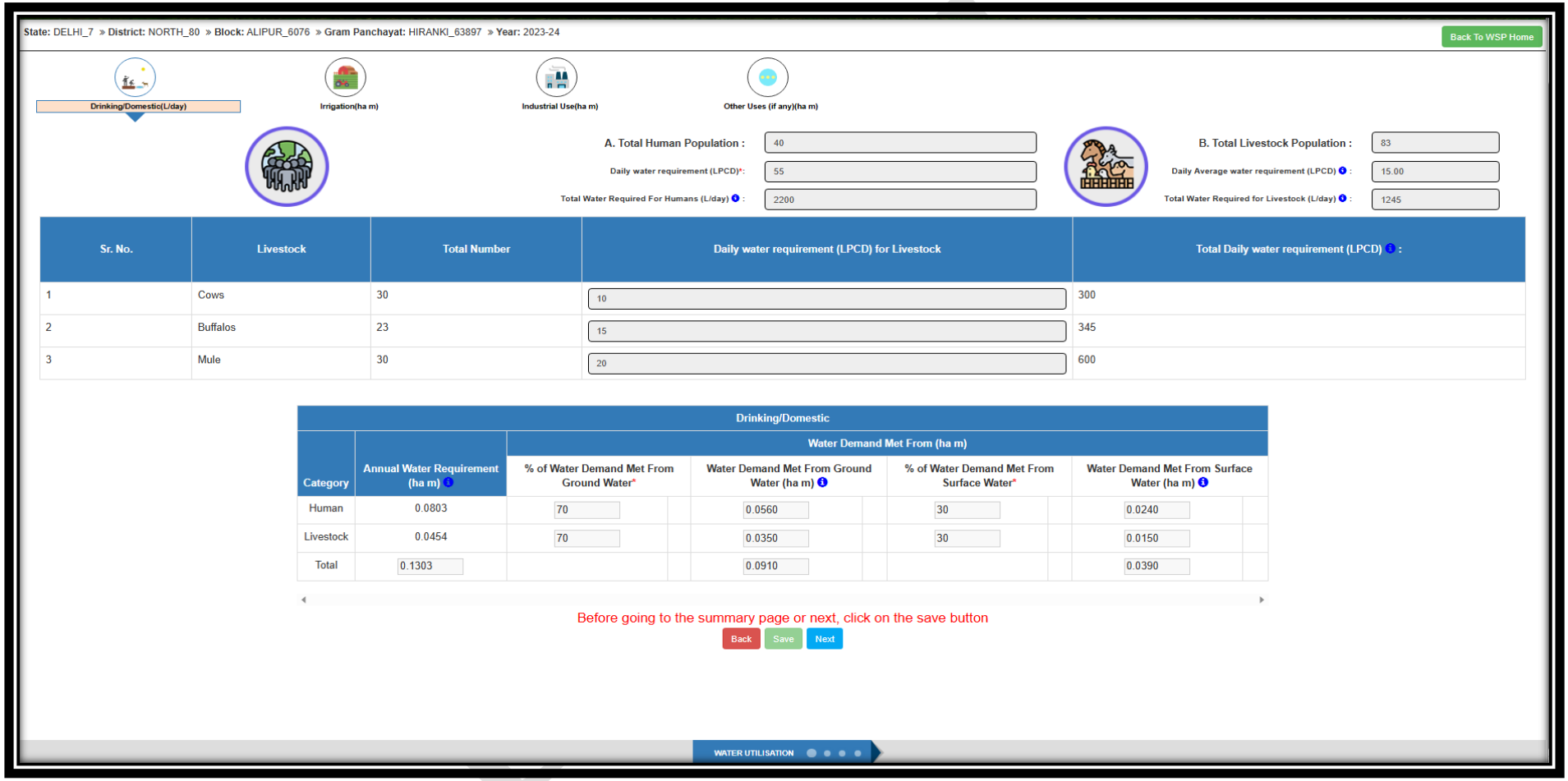


Fig: 3.1

TABLE:3.1

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	**A. Total Human Population**	Auto Fetched from GPDCI module	GPDCI	Auto Fetched	
2	Daily Water Requirement (LPCD)	User Entry	JJM Norms of 55 LPCD or accepted norms of the state	DIP/DPMU	
3	Total Water Required for Humans (L/day)	Auto Calculated		Auto Calculated	Total Human Population * Daily Water Requirement (LPCD)
4	**B. Total Livestock Population**	Details captured from GPDCI module	GPDCI	Auto Fetched	
5	Daily Average Water Requirement (LPCD)	Auto calculated from details entered separate Table given below	Department of Animal Husbandry and Dairy / State Govt. standards	DIP/ DPMU	Total Daily Water requirement/ Total Livestock Population
5 (a)	Livestock and Total Number	Livestock and no. of livestock captured from GPDCI	GPDCI	Auto Fetched	
5 (b)	Daily water requirement (LPCD)	Daily water requirement for particular Livestock to be entered	Department of Animal Husbandry and Dairy	DIP	
6	Total Water Required for Livestock (L/day)	Auto Calculated		Auto Calculated	Total Livestock Population * Average Water Requirement (LPCD)
7	Annual Water Requirement (ha m)	Annual Water Requirement for I) Human II) Livestock		Auto Calculated	**For Human:[**Total Water Requirement for Human (L/Day)*365]/10^{-7} **For Livestock:**[Total Water Requirement for Livestock (L/Day)*365]/10^{-7}
8	% of Water Demand met from Ground Water	User Entry (Allowed between 1-100 in %) or Auto Calculated only if Surface Water Demand is filled	To be decided based on scheme implemented from Surface Water Supply/Field Survey	DIP/ Auto Calculated	

DRAFT

9	Water Demand met from Ground Water (ha m)	Auto Calculated		Auto Calculated	Annual Water Requirement (ha m) * % of Water Demand met from Ground Water/100
10	% of Water Demand met from Surface Water	Auto Calculated or User Entry (Allowed between 1-100 in %) only if Ground Water demand is not filled	To be decided based on scheme implemented from Surface Water Supply/Field Survey	Auto Calculated/DIP	
11	Water Demand met from Surface Water (ha m)	Auto Calculated		Auto Calculated	Annual Water Requirement (ha m) * % of Water Demand met from Surface Water/100

DRAFT

3.2: Irrigation Water Utilization

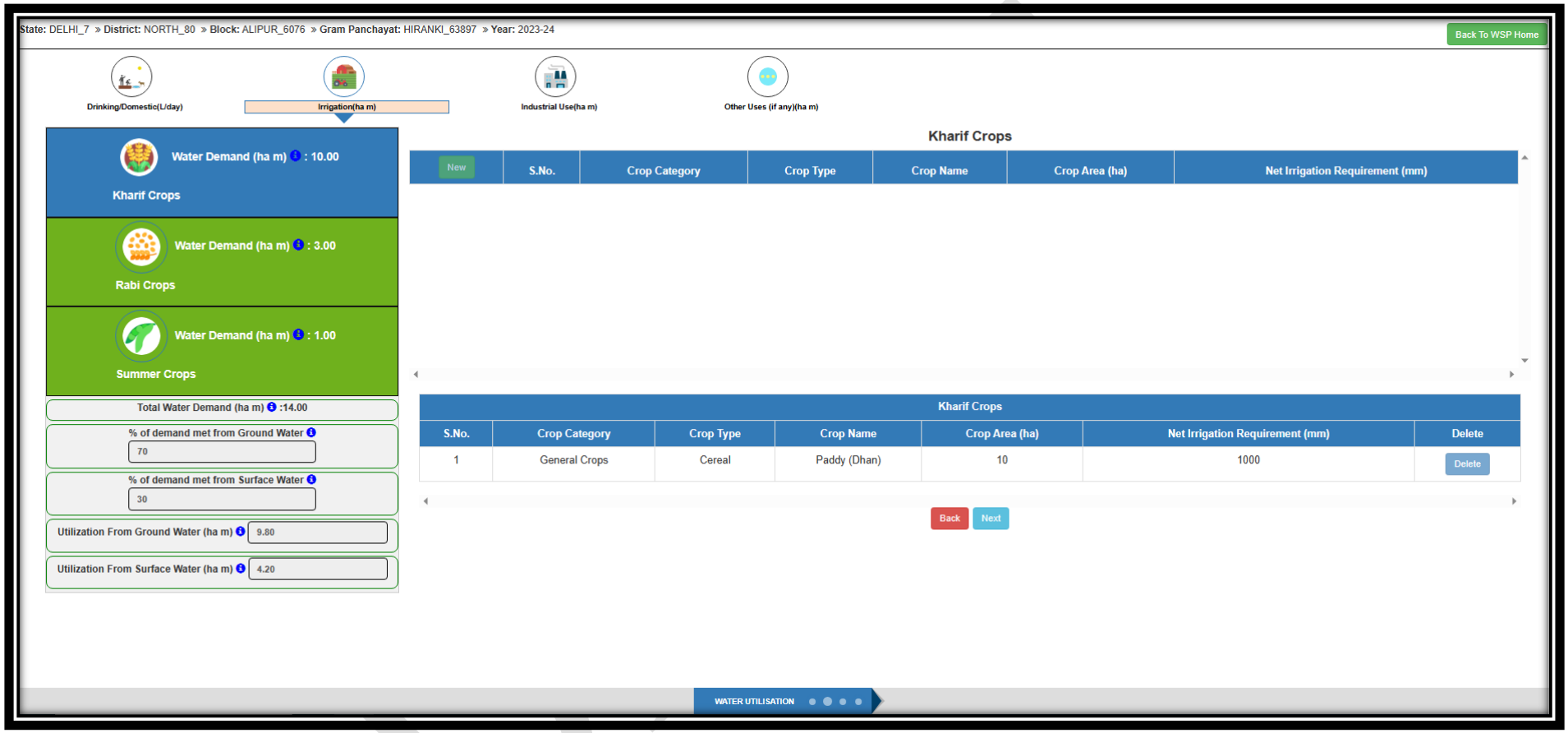


Fig:3.2

TABLE:3.2

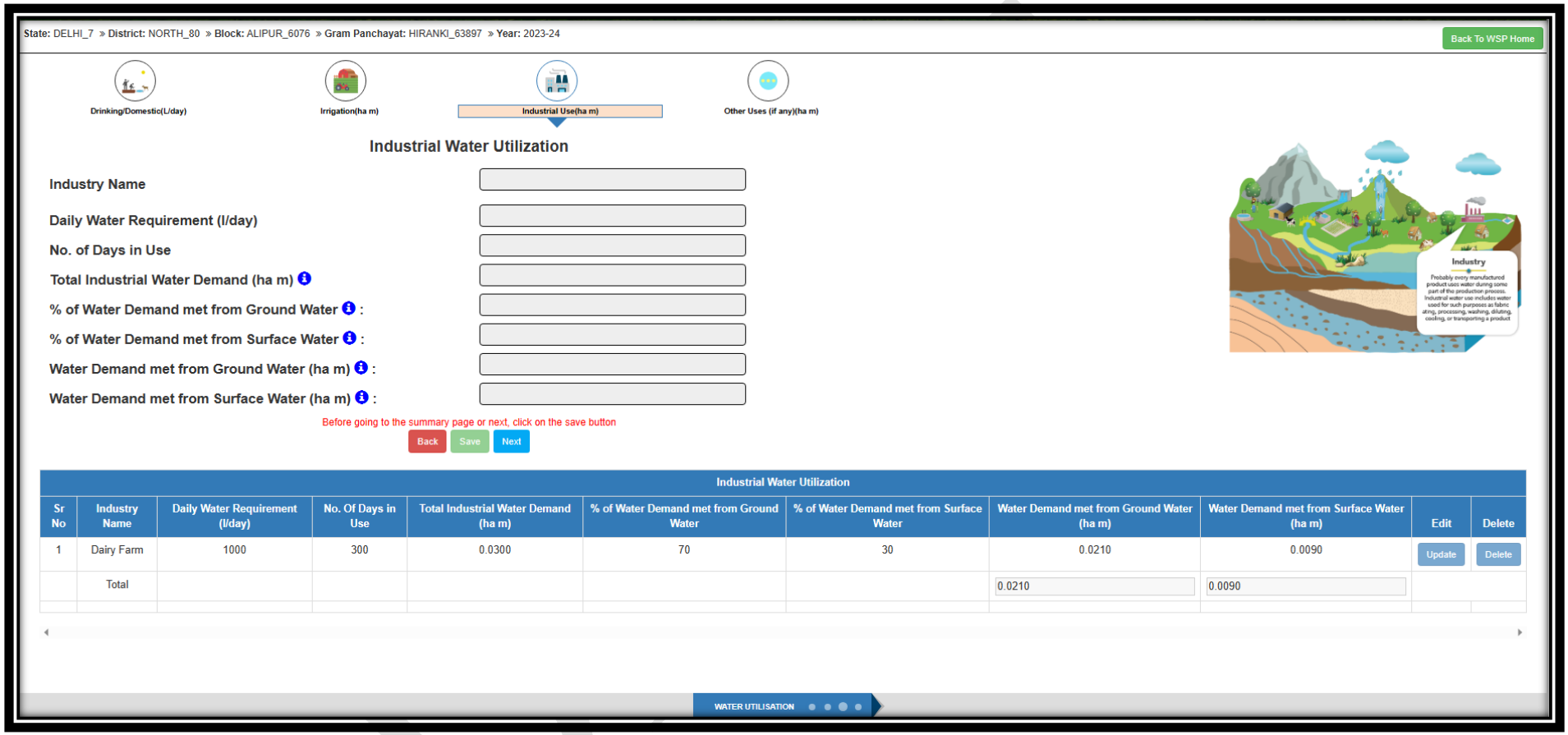
S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	Kharif, Rabi and Summer Crops	Select the Crop category, Crop type, and Crop name from the drop-down	Crop area - Line department	DIP/DPMU	
		NIR(Net Irrigation Requirement) (mm) In case of perennial crops, Put the annual NIR in the planting season. (***Note:*** *It is being proposed to add a separate category for perennial crops in the MIS.*)	• The **Total Crop Water Requirement** will be sourced from the Line Department • **Effective Rainfall** is to be calculated as 70% of the Total seasonal received by a crop. • The Total Rainfall data is to be obtained from IMD/India WRIS.	DIP/DPMU	For Kharif season: NIR=**Total Crop water requirement** of the crop (mm)-**Effective rainfall** received during the cropping period of particular crop (mm). For Rabi and Summer season, whole water requirement of a crop may be considered as NIR.
		Water Demand (ha m)		Auto Calculated	Crop Area (ha) * Net Irrigation Requirement (mm)/1000
		Total Water Demand (ha m)		Auto Calculated	Kharif Crop Water Demand + Rabi Crop Water Demand+ Summer Crop Water Demand
		% of Demand met from Ground Water; User Entry (Allowed between1-100 in %)	To be decided based on scheme implemented from Surface Water Supply/Field Survey	DIP	

DRAFT

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
		% of Demand met from Surface Water		Auto Calculated	100- (% of Demand met from Ground Water)
		Utilization from Ground Water (ha m)		Auto Calculated	Total Ground Water Demand *(% of Demand met from Ground Water/100)
		Utilization from Surface Water (ha m)		Auto Calculated	Total Surface Water Demand *(% of Demand met from Surface Water/100)

DRAFT

3.3: Industrial Water Utilization



Industrial Water Utilization

Sr No	Industry Name	Daily Water Requirement (l/day)	No. Of Days in Use	Total Industrial Water Demand (ha m)	% of Water Demand met from Ground Water	% of Water Demand met from Surface Water	Water Demand met from Ground Water (ha m)	Water Demand met from Surface Water (ha m)	Edit	Delete
1	Dairy Farm	1000	300	0.0300	70	30	0.0210	0.0090	Update	Delete
	Total						0.0210	0.0090		

Fig: 3.3

TABLE: 3.3

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	Industry Name	User Entry	Line Department/Industry	DIP	
2	Daily Water Requirement (L/day)	User Entry	Line Department/Industry	DIP	
3	No. of Days in Use	User Entry	Line Department/Industry	DIP	
4	Total Water Requirement (ha m)	Auto Calculated	Line Department/Industry	Auto calculated	Daily Water Requirement (L/Day) * No. of Days in Use * 0.0000001
5	% of water Demand met from Ground Water	User Entry (Allowed between 1-100%)	Line Department/Industry	DIP	
6	% of water Demand met from Surface Water	Auto Calculated		Auto Calculated	100-(% of Demand met from Ground Water)
7	Water Demand met from Ground water (ha m)	Auto Calculated		Auto Calculated	Total water requirement * (% of water demand met from Ground Water/100)
8	Water Demand met from Surface water (ha m)	Auto Calculated		Auto Calculated	Total water requirement * (% of water demand met from Surface Water/100)

DRAFT

3.4: Other Uses



Fig:3.4

TABLE:3.4

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
i	Type of use	User Entry (Water used other than drinking/domestic, Irrigation, Industrial. For example- Water Park etc.)	Line Department/Industry	DIP	
ii	Daily Water Requirement (L/day)	User Entry	Line Department/Industry	DIP	
iii	No. of Days in Use	User Entry	Line Department/Industry	DIP	
iv	Total Water Requirement (ha m)	Auto Calculated	Line Department/Industry		Daily Water Requirement (L/Day) * No. of Days in Use * 0.0000001
v	% of water Demand met from Ground Water	User Entry (Allowed between 1-100%)	Line Department/Industry	DIP	
vi	% of water Demand met from Surface Water	Auto Calculated		Auto Calculated	100- (% of Demand met from Ground Water)
vii	Water Demand met from Ground water (ha m)	Auto Calculated		Auto Calculated	Total water requirement * (% of water demand met from Ground Water/100)
viii	Water Demand met from Surface water (ha m)	Auto Calculated		Auto Calculated	Total water requirement * (% of water demand met from Surface Water/100)

DRAFT

4. WATER BALANCE SUB MODULE

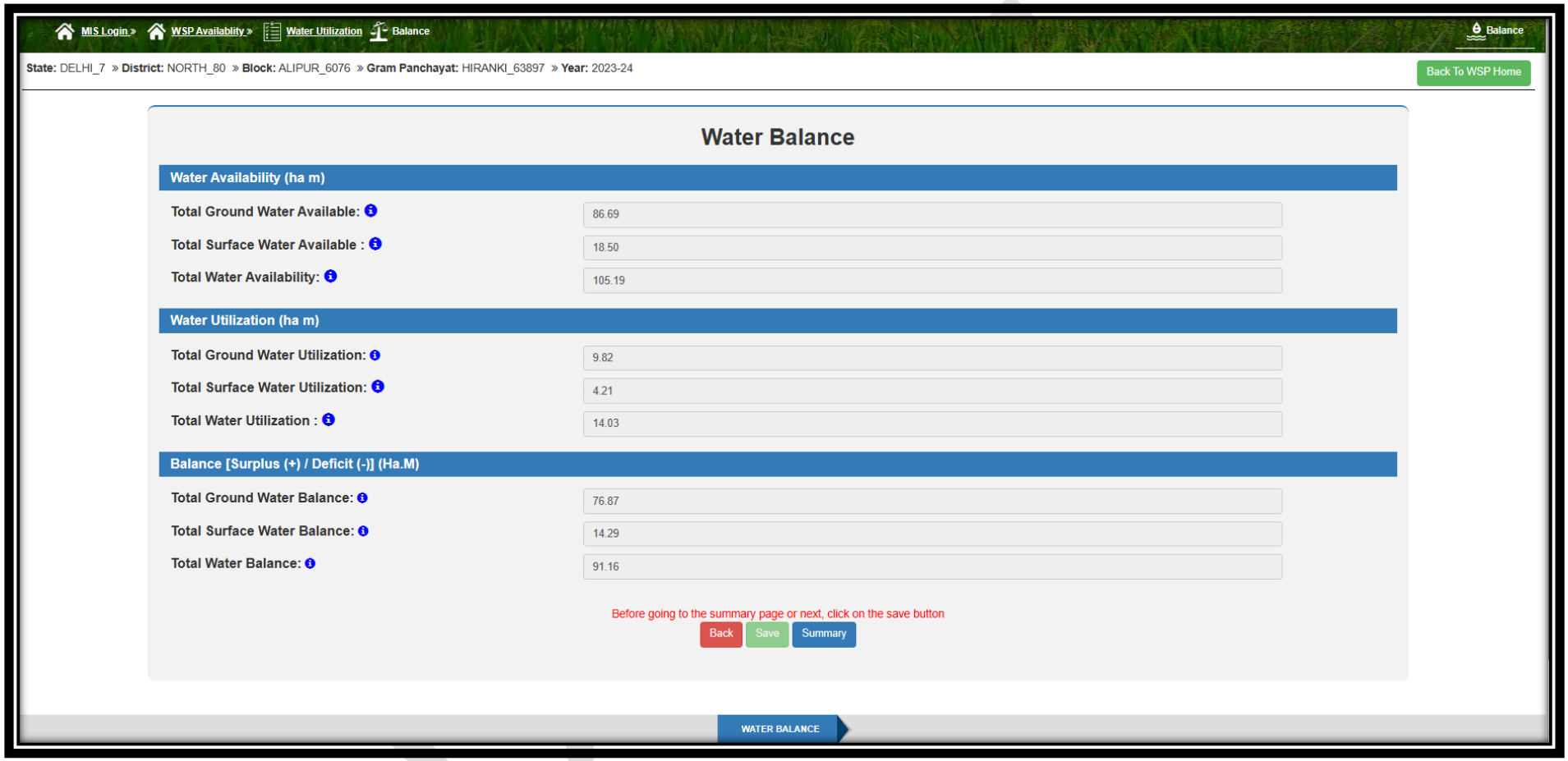


Fig:4

TABLE: 4

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	Total Ground Water Available (ha m)	Auto Fetched	Water Availability Sub Module from Point No.2 Ground Water f. Total Ground Water Available (ha m)	Auto Calculated	
2	Total Surface Water Available (ha m)	Auto Fetched	Water Availability Sub Module from Point No. 3 Surface Water c. Total Surface Water Available (ha m)	Auto Calculated	
3	Total Water Availability (ha m)	Auto Calculated		Auto Calculated	Total Ground Water Availability (ha m) + Total Surface Water Available (ha m)
4	Total Ground Water Utilization (ha m)	Auto Fetched	Water Utilization Module	Auto Calculated	[Drinking/Domestic-Water Demand met from Ground Water (ha m)] + [Irrigation-Water Demand met from Ground Water (ha m)] + [Industrial-Water Demand met from Ground Water (ha m)] + [Other -Water Demand met from Ground Water (ha m)]
5	Total Surface Water Utilization (ha m)	Auto Fetched	Water Utilization Module	Auto Calculated	[Drinking/Domestic-Water Demand met from Surface Water (ha m)] + [Irrigation-Water Demand met from Surface Water (ha m)] + [Industrial-Water Demand met from Surface Water (ha m)] + [Other -Water Demand met from Surface Water (ha m)]
6	Total Water Utilization (ha m)	Auto Calculated		Auto Calculated	Utilization of Ground Water (ha m) + Utilization of Surface Water (ha m)
7	Total Ground Water Balance (ha m)	Auto Calculated		Auto Calculated	Total Ground Water Available (ha m)- Total Ground Water Utilization (ha m)
8	Total Surface Water Balance (ha m)	Auto Calculated		Auto Calculated	Total Surface Water Available (ha m) - Total Surface Water Utilization (ha m)
9	Total Water Balance (ha m)	Surplus (+)/ Deficit (-)		Auto Calculated	Total Ground Water Balance (ha m) + Total Surface Water Balance (ha m)

DRAFT

5. WATER BUDGET SUB MODULE

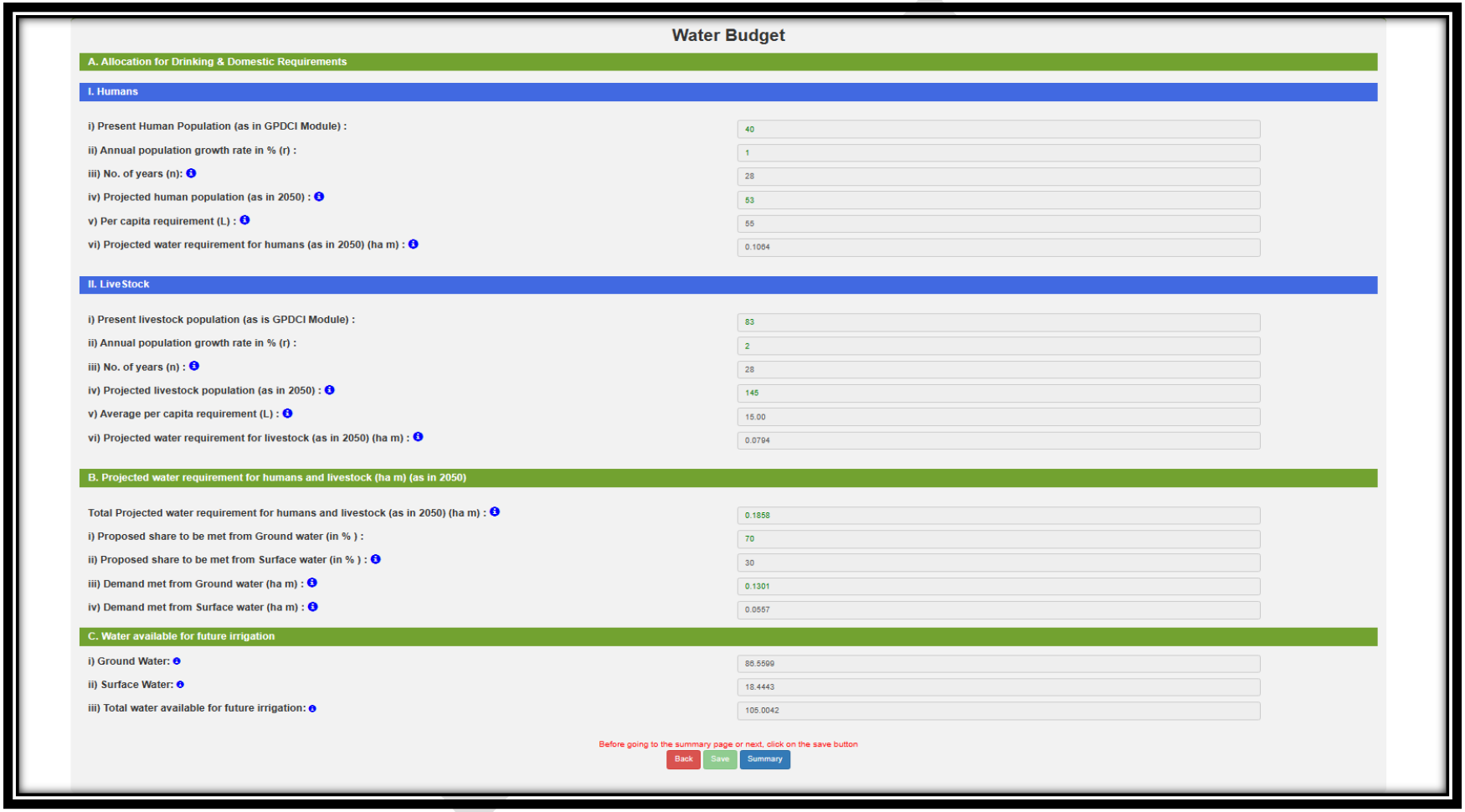


Fig:5

TABLE:5

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	A. I. (i)	Present Human Population; Detail captured from GPDCI module	GPDCI	DIP	
2	A. I. (ii)	Annual Population Growth Rate (r) in % to be entered	Census / any standard document		
3	A. I. (iii)	No. of years (n) ; Detail captured from GPDCI module	GPDCI	DIP	2050-year for which the present population was entered in GPDCI Module
4	A. I. (iv)	Projected Human Population (as in 2050)		Auto Calculated	Present human Population*$(1+r)^n$
5	A. I. (v)	Per Capita Requirement (L)	Water Utilization Module: Daily Water Requirement (LPCD)	Auto Fetched	
6	A. I. (vi)	Projected water requirement for humans (ha m) (as in 2050)		Auto Calculated	Projected Human Population (as in 2050)* per capita requirement*365*10^{-7}
7	A. II. (i)	Present Livestock Population; Detail captured from GPDCI module	GPDCI	DIP	
8	A. II. (ii)	Annual Population Growth Rate (r) in % to be entered	Department of Animal Husbandry and Dairy		
9	A. II. (iii)	No. of years (n) ; Detail captured from GPDCI module	GPDCI	DIP	2050-year for which the present population was entered in GPDCI Module
10	A. II. (iv)	Projected Livestock Population (as in 2050)		Auto Calculated	Present Livestock Population*$(1+r)^n$
11	A. II. (v)	Average Per Capita Requirement (L)	Water Utilization Module: Average Water Requirement (LPCD)	Auto Fetched	
12	A. II. (vi)	Projected water requirement for livestock (ha m)		Auto Calculated	Projected livestock Population (as in 2050)* Average per capita requirement*365*10^{-7}

DRAFT

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
13	B. Total Projected water requirement for humans and livestock (as in 2050) (ha m)	Auto Calculated		Auto Calculated	Projected water requirement for humans (as in 2050) + Projected water requirement for livestock (as in 2050)
14	B. (i)	Proposed share to be met from Ground Water; % to be entered		DIP	
15	B. (ii)	Proposed share to be met from Surface Water		Auto Calculated	100-Proposed share (in %) to be met from Ground Water
16	B. (iii)	Demand met from Ground Water (ha m)		Auto Calculated	Projected water requirement for humans and livestock (as in 2050)*(Proposed share (in %) to be met from Ground water/100)
17	B. (iv)	Demand met from Surface Water (ha m)		Auto Calculated	Projected water requirement for humans and livestock (as in 2050)*(Proposed share (in %) to be met from Surface water/100)
18	C. (i)	Water available for future Irrigation (Ground Water) (ha m)	Total Ground Water Available : From Water Availability (2 f) Demand met from Ground Water : From Water Budget (B-iii)	Auto Calculated	Total Ground Water Available (ha m) - Demand met from Ground Water (ha m)
19	C. (ii)	Water available for future Irrigation (Surface Water) (ha m)	Total Surface Water Available : From Water Availability (3 c) Demand met from Surface Water : From Water Budget (B-iv)	Auto Calculated	Total Surface Water Available (ha m) - Demand met from Surface Water (ha m)
20	C. (iii)	Total Water available for future irrigation (ha m)		Auto Calculated	Ground Water (ha m) +Surface Water (ha m)

DRAFT

6. DEMAND SIDE SUB MODULE

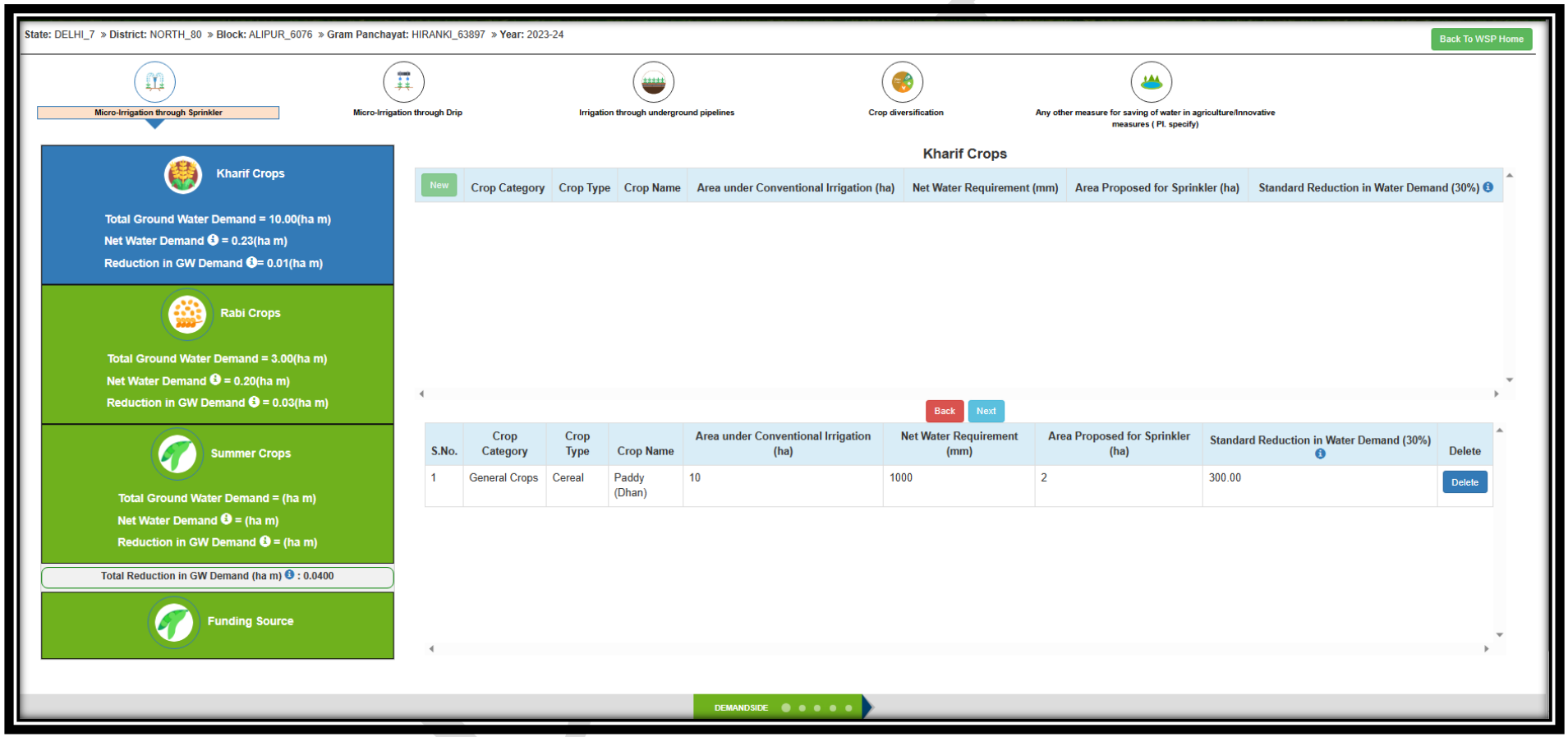


Fig: 6.1

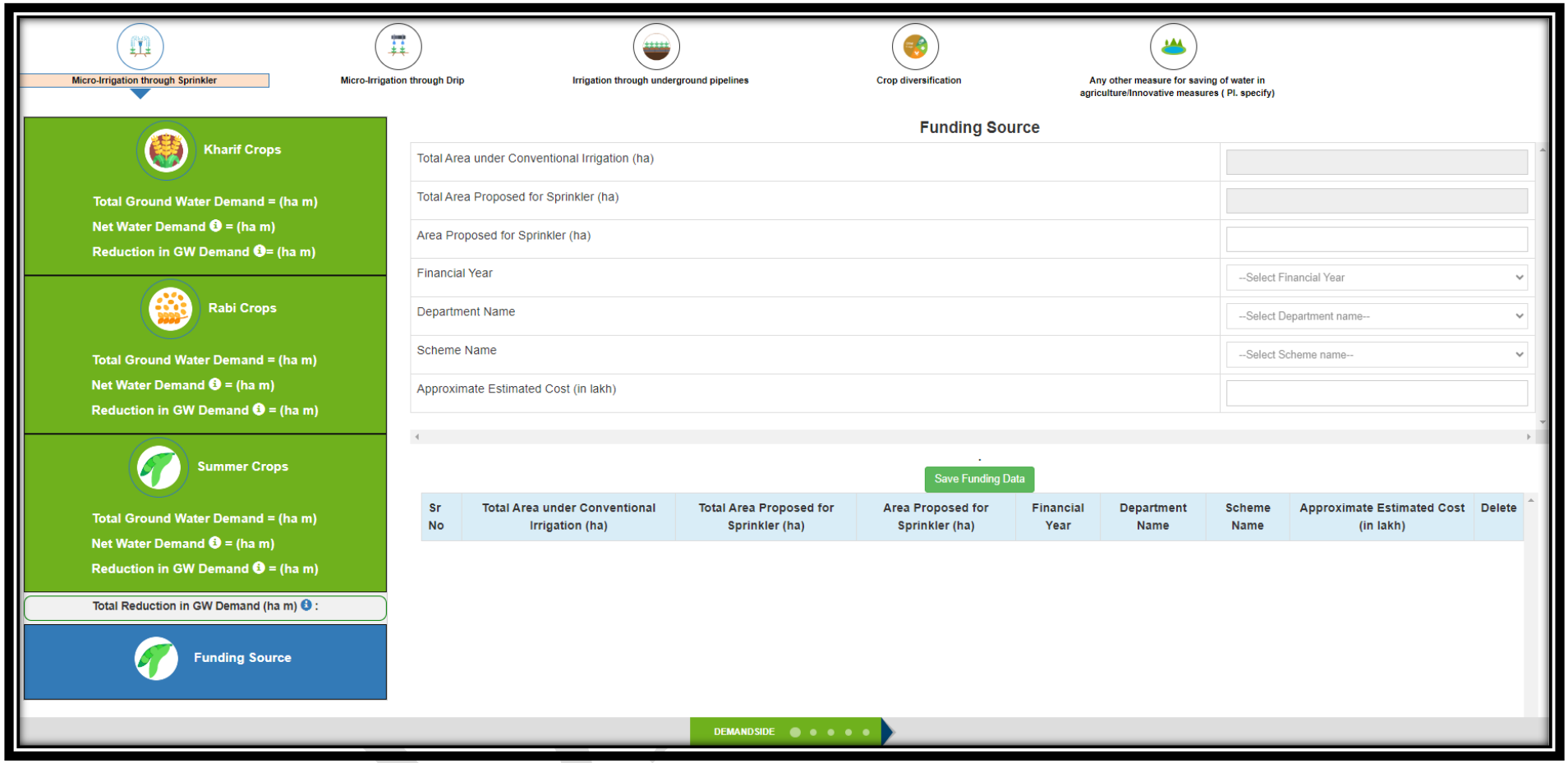

Micro-Irrigation through Sprinkler
Micro-Irrigation through Drip
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture/Innovative measures (Pl. specify)
Kharif Crops
Total Ground Water Demand = (ha m)
Net Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Rabi Crops
Total Ground Water Demand = (ha m)
Net Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Summer Crops
Total Ground Water Demand = (ha m)
Net Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Total Reduction in GW Demand (ha m) :
Funding Source
Funding Source
Total Area under Conventional Irrigation (ha)
Total Area Proposed for Sprinkler (ha)
Area Proposed for Sprinkler (ha)
Financial Year
--Select Financial Year
Department Name
--Select Department name--
Scheme Name
--Select Scheme name--
Approximate Estimated Cost (in lakh)
Save Funding Data
Sr No
Total Area under Conventional Irrigation (ha)
Total Area Proposed for Sprinkler (ha)
Area Proposed for Sprinkler (ha)
Financial Year
Department Name
Scheme Name
Approximate Estimated Cost (in lakh)
Delete
DEMANDSIDE

Fig:6.2

TABLE: 6.1

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1.	**Micro Irrigation through Sprinkler**				
	(Kharif , Rabi, and Summer Crops)	Choose the identical Crop category, Crop type and Crop name from the drop-down menu as the ones filled in the Water Utilization Sub Module.	Water Utilization Sub Module	DIP	
		Area under Conventional Irrigation (ha) ; Input the same Area value as previously filled in the Water Utilization Sub Module.	Water Utilization Sub Module	DIP	
		Net Water Requirement (mm); Kindly input the same NIR value as previously filled in the Water Utilization Sub Module.	Water Utilization Sub Module	DIP	
		Area Proposed for Sprinkler (ha) Note: *In the case of perennial crops, whole area under sprinkler may be proposed in the planting season*	Area proposed by communities for the particular crop	DIP	
		Standard reduction in Water Demand (30%) (ha m)		Auto Calculated	Water Requirement (mm) *30/100
		Total Ground Water Demand (ha m)		Auto Calculated	Sum of Water Demand for different Crop
		Net Water Demand (ha m)		Auto Calculated	[(Area – Area proposed for Sprinkler through Irrigation) * Water Requirement*0.001] + [Area proposed for Sprinkler Irrigation * Water Requirement (Net Water Requirement – Standard Reduction in Water Demand)]*0.001
		Reduction in GW Demand(ha m)		Auto Calculated	Water Demand (ha m) - Net Water Demand (ha m)
		Total Reduction in GW Demand (ha m)		Auto Calculated	Reduction in GW Demand in Kharif (ha m)+ Reduction in GW Demand in Rabi (ha m)+ Reduction in GW Demand in Summer (ha m)

DRAFT

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
	Funding Source				
	Total Area under conventional Irrigation (Ha)	Detail captured from Water Utilization	Line Department	DIP	
	Total Area Proposed for Sprinkler (Ha)	Auto Fetched from Area proposed under Sprinkler in different season	Area proposed by communities	DIP	
	Area Proposed for Sprinkler (Ha)	Area proposed under different Department /Scheme/F.Y.		DIP	
	Department Name & Scheme Name	Tentative Name of Department & Scheme under which Sprinkler to be provided		DIP	
	Approximate Estimated Cost (Lakh)	User Entry	Schedule of Rate (SOR) of State	DIP	
2.	**Micro-irrigation through Drip**				
	(Kharif , Rabi, and Summer Crops)	All Parameters entered same as explained in S.No. 1 Note: Standard Reduction in Water Demand due to implementation of Drip is 40%			Standard Reduction in Water Demand= Water Requirement (mm) *40/100
	Funding Source	Same as explained in S.No. 1			
3.	**Irrigation through Underground Pipelines**				
	(Kharif , Rabi, and Summer Crops)	All Parameters entered same as explained in S.No. 1 except Standard Reduction in Water Demand which is 20% due to implementation of Underground Pipelines			Standard Reduction in Water Demand= Water Requirement (mm) *20/100
	Funding Source	Same as explained in S.No. 1			

DRAFT

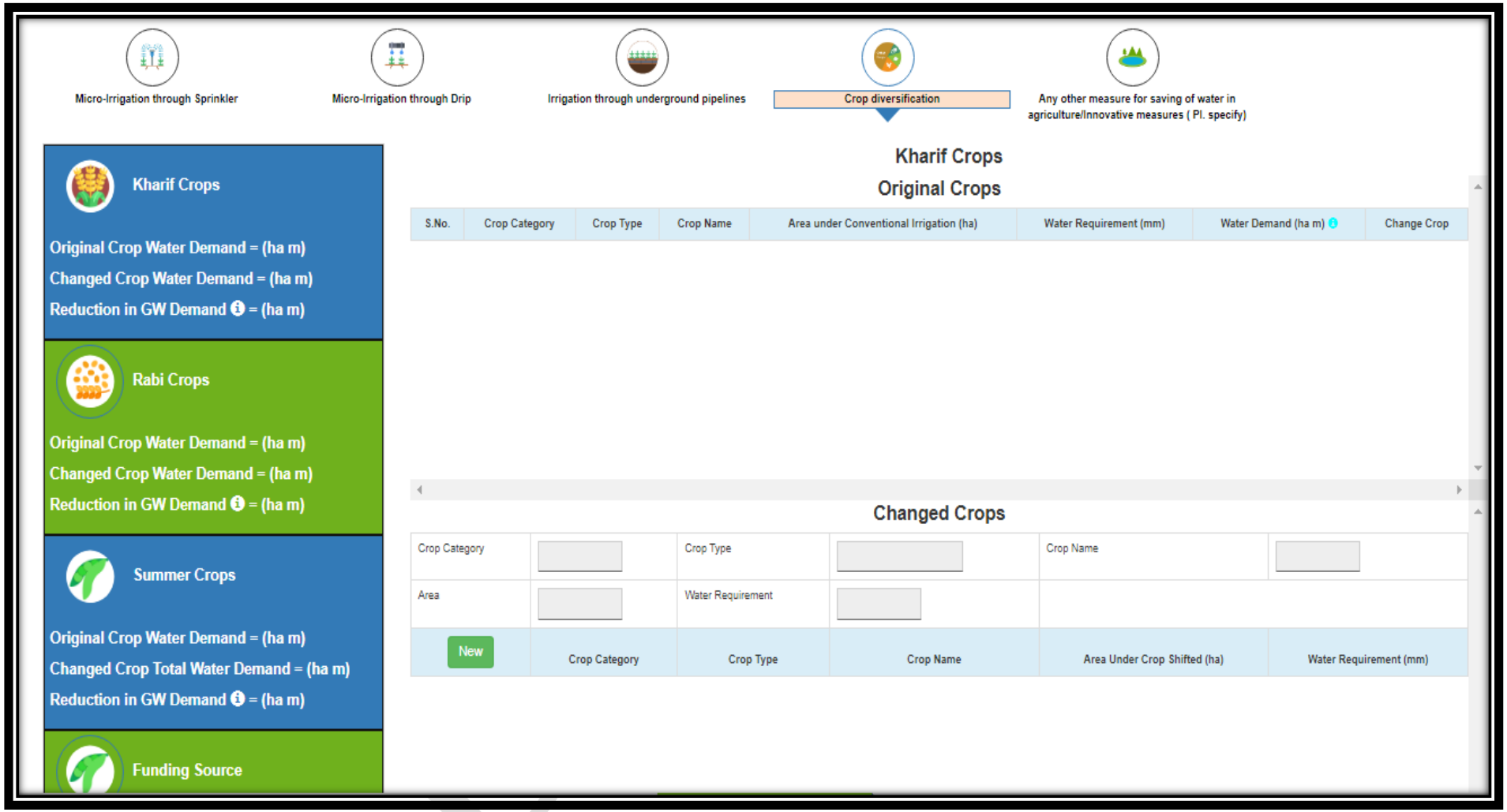

Micro-Irrigation through Sprinkler
Micro-Irrigation through Drip
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture/Innovative measures (Pl. specify)
Kharif Crops
Original Crop Water Demand = (ha m)
Changed Crop Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Rabi Crops
Original Crop Water Demand = (ha m)
Changed Crop Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Summer Crops
Original Crop Water Demand = (ha m)
Changed Crop Total Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Funding Source
Kharif Crops
Original Crops
S.No.
Crop Category
Crop Type
Crop Name
Area under Conventional Irrigation (ha)
Water Requirement (mm)
Water Demand (ha m)
Change Crop
Changed Crops
Crop Category
Crop Type
Crop Name
Area
Water Requirement
New
Crop Category
Crop Type
Crop Name
Area Under Crop Shifted (ha)
Water Requirement (mm)

Fig: 6.3

Table: 6.2

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
4.	**Crop Diversification**				
	(Kharif , Rabi, and Summer Crops)	**Original Crops:** Crop Category, Crop Type , Crop Name , Area under Conventional Irrigation and Water Requirement ; Details captured from Water Utilization Sub Module	Water Utilization Sub Module	Auto Fetched	
		Original Crops: Water Demand (ha m)		Auto Calculated	Area (ha) * Water Requirement (mm)/1000
		Changed Crops : Low Water consuming Crop Category, Crop Type , Crop Name to be selected from Drop down	Community Meeting	DIP	
		Area under shifted crop to be entered (ha)	Area proposed by communities	DIP	
		Water Requirement for the changed crop to be entered (mm)	State Govt	DIP	
		Original Crop Water Demand (ha m)		Auto Calculated	
		Changed Crop Water Demand (ha m)		Auto Calculated	
		Reduction in GW Demand (ha m)		Auto Calculated	Original Crop Water Demand (ha m) - Changed Crop Water Demand (ha m)
		Total Reduction in GW Demand (ha m)		Auto Calculated	Reduction in GW Demand Kharif (ha m)+ Reduction in GW Demand Rabi (ha m)+ Reduction in GW Demand Summer (ha m)
	Funding Source	Same as explained in S.No. 1			

DRAFT

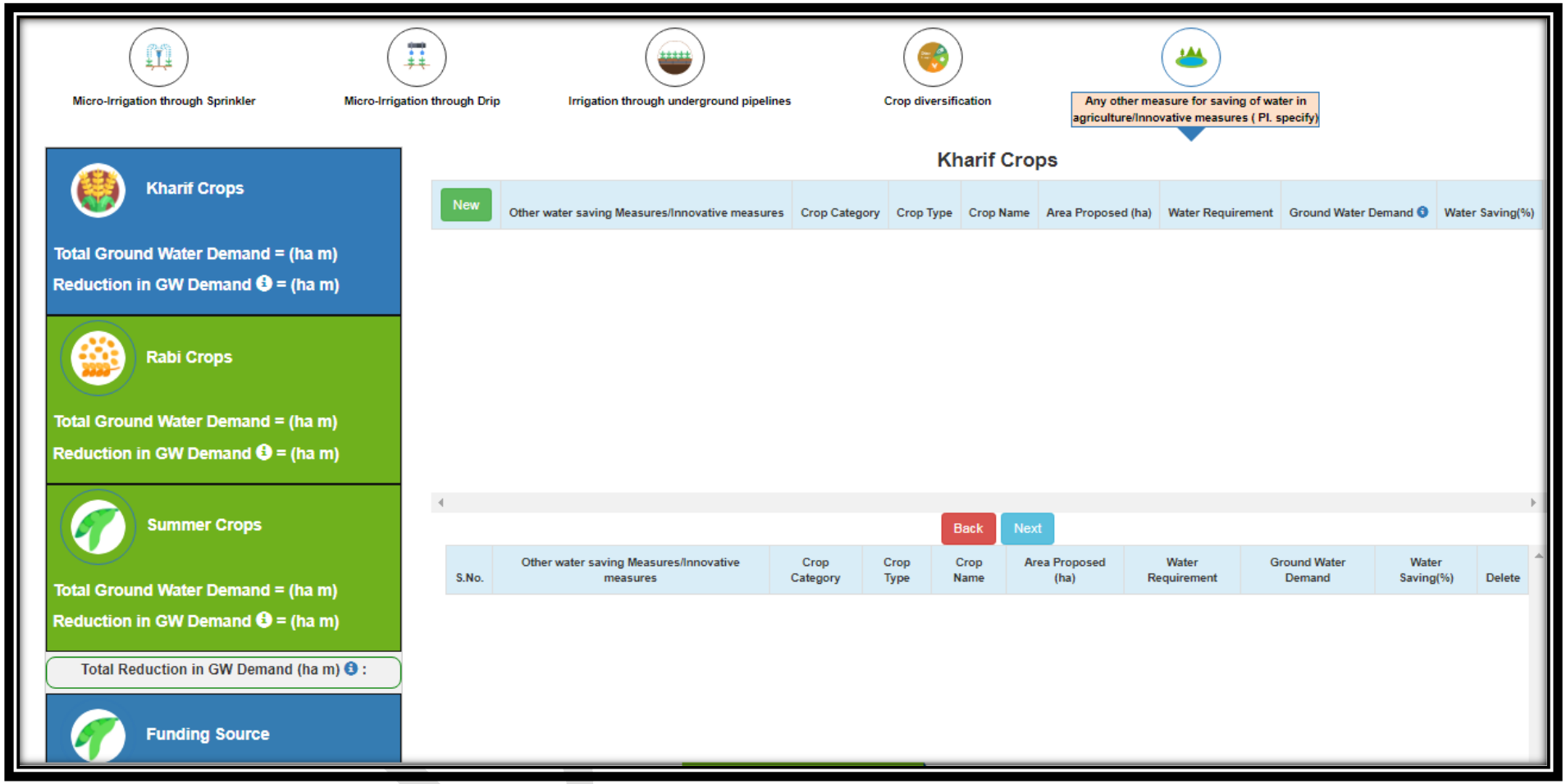

Micro-Irrigation through Sprinkler
Micro-Irrigation through Drip
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture/Innovative measures (Pl. specify)
Kharif Crops
Total Ground Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Rabi Crops
Total Ground Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Summer Crops
Total Ground Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Total Reduction in GW Demand (ha m) :
Funding Source
Kharif Crops
New
Other water saving Measures/Innovative measures
Crop Category
Crop Type
Crop Name
Area Proposed (ha)
Water Requirement
Ground Water Demand
Water Saving(%)
Back
Next
S.No.
Other water saving Measures/Innovative measures
Crop Category
Crop Type
Crop Name
Area Proposed (ha)
Water Requirement
Ground Water Demand
Water Saving(%)
Delete

Fig: 6.4

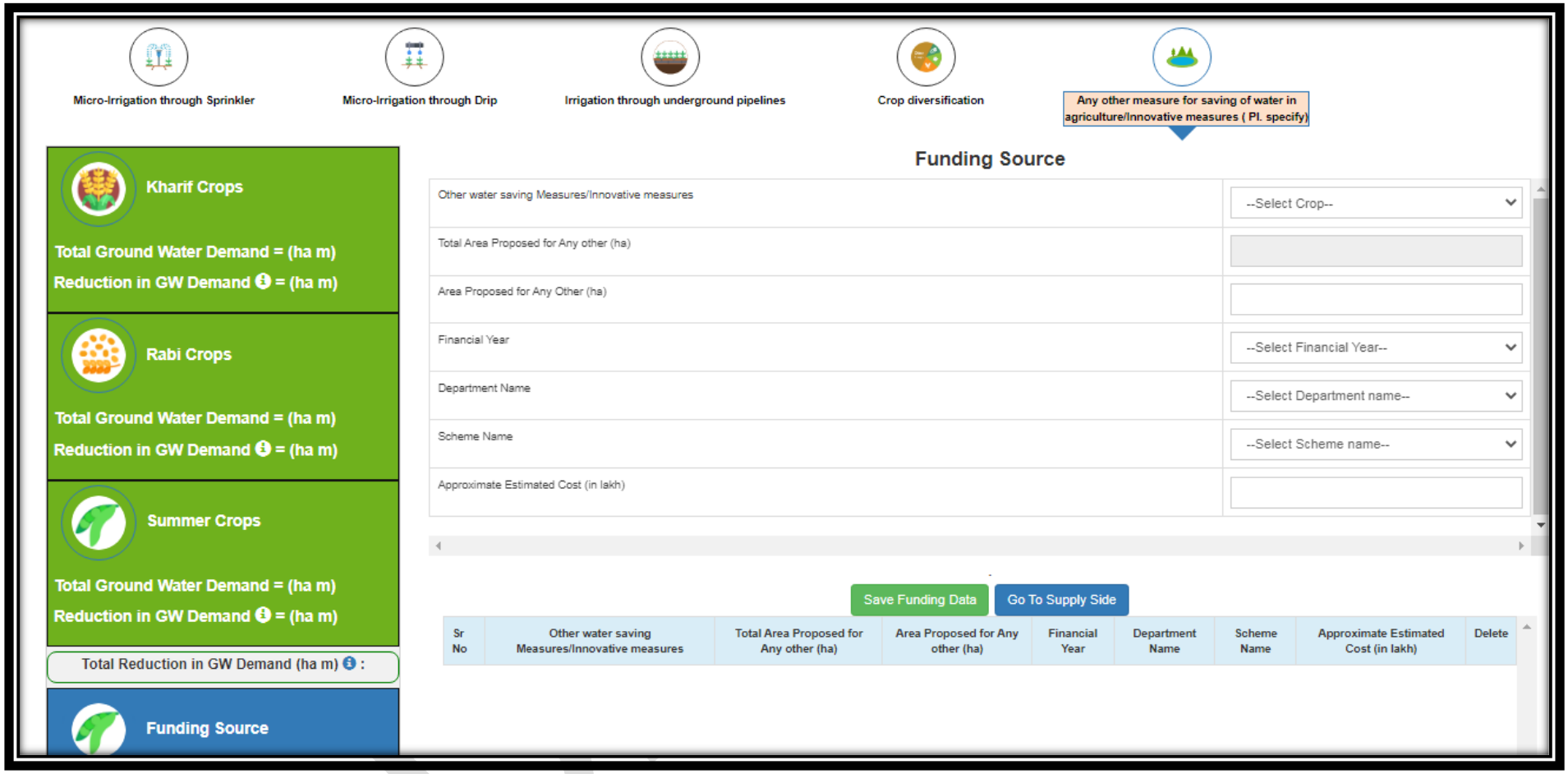

Micro-Irrigation through Sprinkler
Micro-Irrigation through Drip
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture/Innovative measures (Pl. specify)
Kharif Crops
Total Ground Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Rabi Crops
Total Ground Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Summer Crops
Total Ground Water Demand = (ha m)
Reduction in GW Demand = (ha m)
Total Reduction in GW Demand (ha m) :
Funding Source
Funding Source
Other water saving Measures/Innovative measures
--Select Crop--
Total Area Proposed for Any other (ha)
Area Proposed for Any Other (ha)
Financial Year
--Select Financial Year--
Department Name
--Select Department name--
Scheme Name
--Select Scheme name--
Approximate Estimated Cost (in lakh)
Save Funding Data
Go To Supply Side
Sr No
Other water saving Measures/Innovative measures
Total Area Proposed for Any other (ha)
Area Proposed for Any other (ha)
Financial Year
Department Name
Scheme Name
Approximate Estimated Cost (in lakh)
Delete

Fig: 6.5

Table: 6.3

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
5.	**Any other measure for saving of water in agriculture/innovative measures**				
	(Kharif , Rabi, and Summer Crops)	Select the Other water saving Measures/Innovative measures, Crop category, Crop type and crop name from drop down		DIP	
		Area Proposed (ha) to be entered	Area proposed by communities	DIP	
		Water Requirement to be entered	State Govt.	DIP	
		Ground Water Demand (ha m)		Auto Calculated	Area proposed in Ha *Water Requirement/1000
		Water Saving (%) to be entered	Line Department	DIP	
		Total Ground Water Demand (ha m)		Auto Calculated	Sum of Water Demand for different Crop
		Reduction in GW Demand (ha m)		Auto Calculated	Ground Water Demand (ha m)* Water Saving * 0.01
		Total Reduction in GW Demand (ha m)		Auto Calculated	Reduction in GW Demand Kharif (ha m)+ Reduction in GW Demand Rabi (ha m)+ Reduction in GW Demand Summer (ha m)
	Funding Source				
	Other Water Saving Measures/Innovative Measures	Select the Other water saving Measures/Innovative measures which has been selected earlier	Any other measure for saving of water in agriculture/innovative measures for Kharif, Rabi and Summer Crops	DIP	
	Total Area Proposed under any other (Ha)	Auto Fetched from Area proposed under any other in different season	Area proposed by communities	DIP	
	Area Proposed for any other (Ha)	Area proposed under different Department or Scheme		DIP	
	Department Name & Scheme Name	Tentative Name of Department & Scheme		DIP	
	Approximate Estimated Cost (Lakh)	User Entry	Schedule of Rate (SOR) of State	DIP	

DRAFT

7. SUPPLY SIDE SUB MODULE

7.1 Artificial Recharge Structure

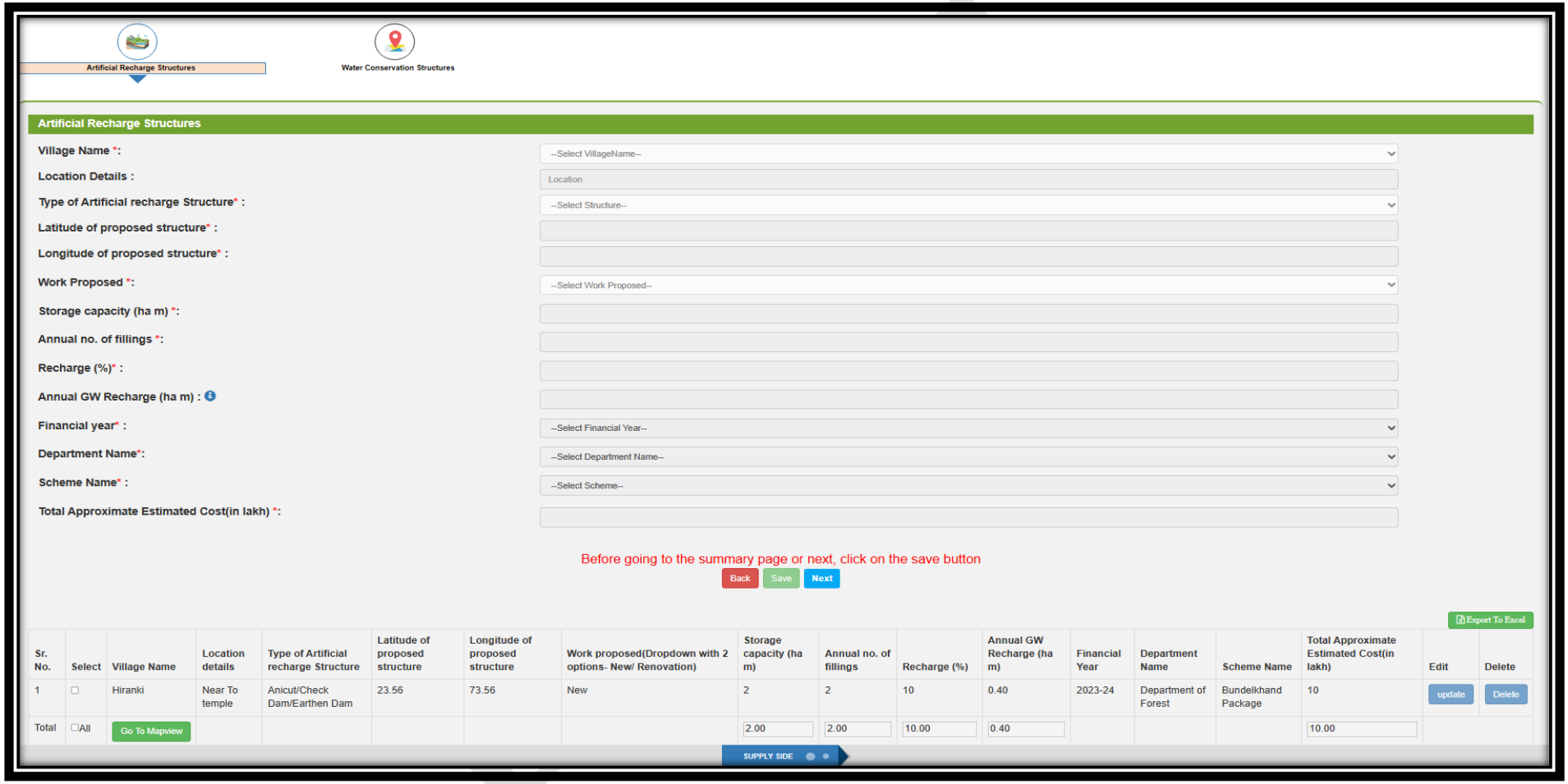


Fig:7.1

TABLE: 7.1

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	Village Name	Select from Dropdown	Atal Jal Gram Panchayat Master list	DIP	
2	Location Details	Location details where ARS is being Proposed	Line Department/ decided during community meeting	DIP	
3	Type of Artificial Recharge Structure	Select from Dropdown	Line Department/ decided during community meeting	DIP	
4	Latitude and Longitude of Proposed Structure	Coordinates of the ARS	Line Department/ decided during community meeting	DIP	
5	Work Proposed	Select from Dropdown New/ Renovation	Line Department/ decided during community meeting	DIP	
6	Storage Capacity (ha m)	User Entry	As per the proposed design/dimension of structure	DIP	
7	Annual no. of fillings	User Entry	Based on rainfall	DIP	
8	Recharge (%)	User Entry	As per GEC/State norms	DIP	
9	Annual GW Recharge (ha m)	Auto Calculated			Storage Capacity (ha m) *Annual no. of fillings*Recharge %*0.01
10	Financial Year	Select from Dropdown	Line Department	DIP	
11	Department Name	Select from Dropdown	Line Department	DIP	
12	Scheme Name	Select from Dropdown	Line Department	DIP	
13	Total Approximate Estimated Cost (in Lakh)	User Entry	Schedule of Rate (SOR) of State	DIP	

DRAFT

7.2 Water Conservation Structure

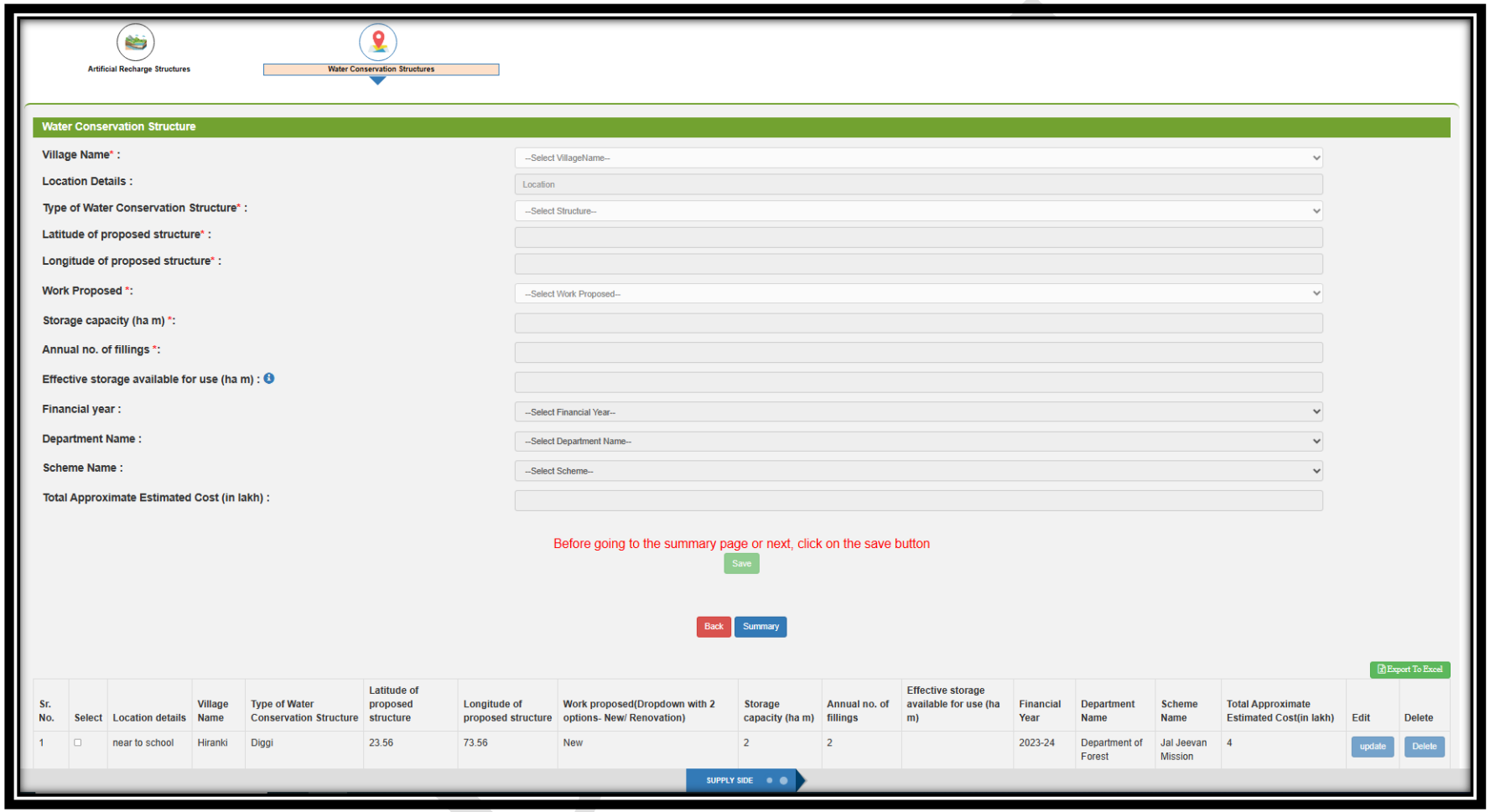


Fig: 7.2

TABLE: 7.2

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	Village Name	Select from Dropdown	Atal Jal Gram Panchayat Master list	DIP	
2	Location Details	Location details where WCS is being Proposed	Line Department/ decided during community meeting	DIP	
3	Type of Water Conservation Structure	Select from Dropdown	Line Department/ decided during community meeting	DIP	
4	Latitude and Longitude of Proposed Structure	Coordinates of the WCS	Line Department/ decided during community meeting	DIP	
5	Work Proposed	Select from Dropdown	Line Department/ decided during community meeting	DIP	
6	Storage Capacity (ha m)	User Entry	As per the proposed design/dimension of structure	DIP	
7	Annual no. of fillings	User Entry	Based on rainfall	DIP	
8	Effective Storage Available for use (ha m)	Auto Calculated			Storage Capacity*0.6
9	Financial Year	Select from Dropdown	Line Department	DIP	
10	Department Name	Select from Dropdown	Line Department	DIP	
11	Scheme Name	Select from Dropdown	Line Department	DIP	
12	Total Approximate Estimated Cost (in Lakh)	User Entry	SOR of the State/Line Department	DIP	

DRAFT

8. PGWM COMMITTEE SUB MODULE

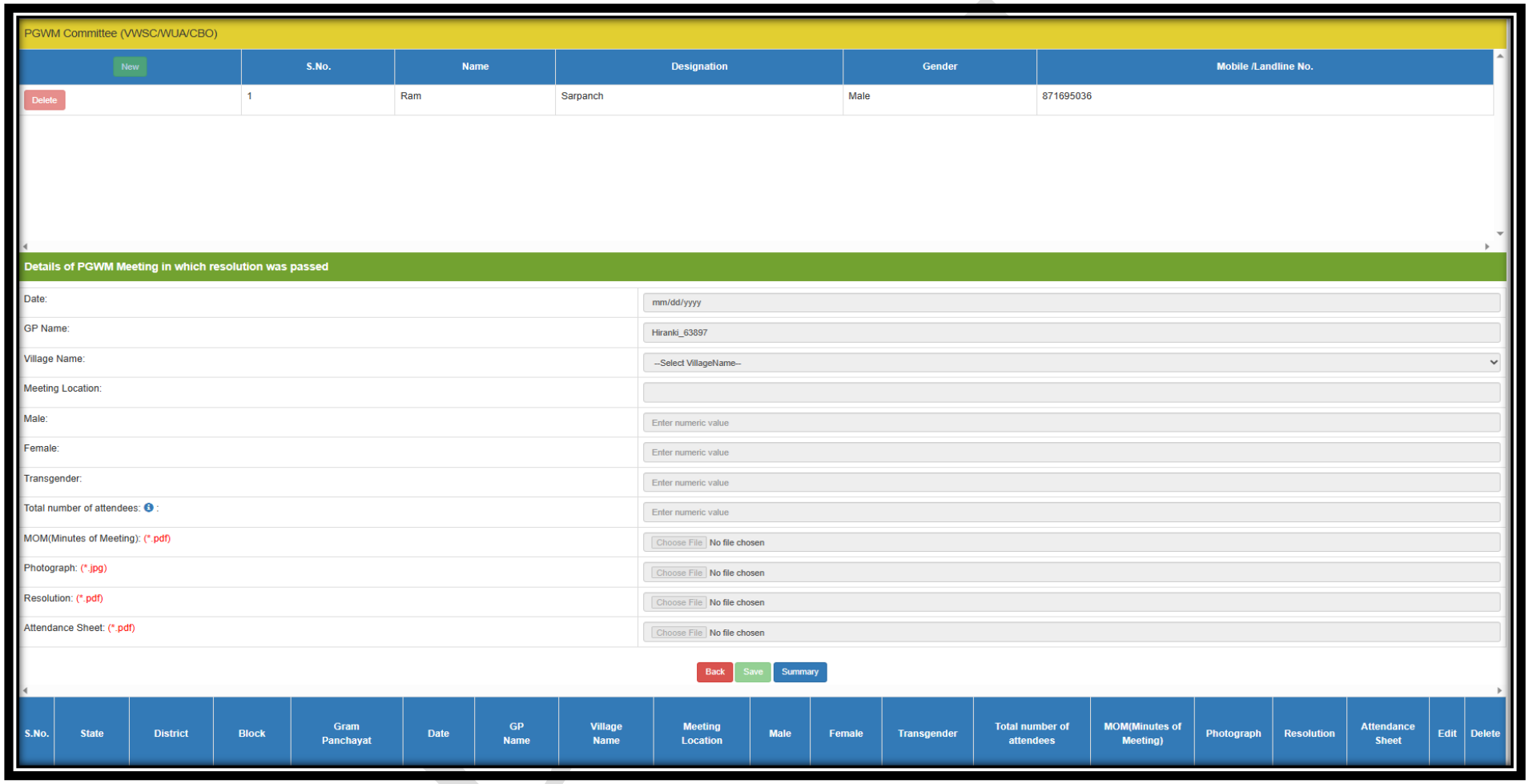


Fig: 8

TABLE: 8

S.No.	Template Item Nos./ attribute	Explanation	Source of Data	By whom	Formula
1	**PGWM Committee (VWSC/WUA/CBO)**				
i	Name, Designation, Gender, Mobile/Landline No.	Details of PGWM committee member to be entered	Gram Panchayat PGWM Committee	DIP	
2	**Details of PGWM Meeting in which resolution was passed**				
i	Date	Date of meeting conducted	PGWM Meeting	DIP	
ii	GP Name	Auto Fetched	PGWM Meeting		
iii	Village Name	Select from Dropdown	Atal Jal Gram Panchayat Master list	DIP	
iv	Meeting Location	Location details of where meeting conducted	PGWM Meeting	DIP	
v	Male, Female, Transgender	No. of Male , Female, Transgender who attended the meeting	PGWM Meeting	DIP	
vi	Total number of attendees	Auto Calculated	PGWM Meeting	Auto Calculated	Male +Female +Transgender
vii	MOM, Photographs, Resolution, Attendance Sheet	Scanned copy of MOM, Resolution passed by community during meeting and attendance sheet in PDF format & Photograph of meeting in JPG format to be uploaded	Upload	DIP	

DRAFT

9. WSP REMARKS

- In case DPMU/SPMU intends to reject the WSP, they have to enter their module-wise remarks here to facilitate modification/corrections.

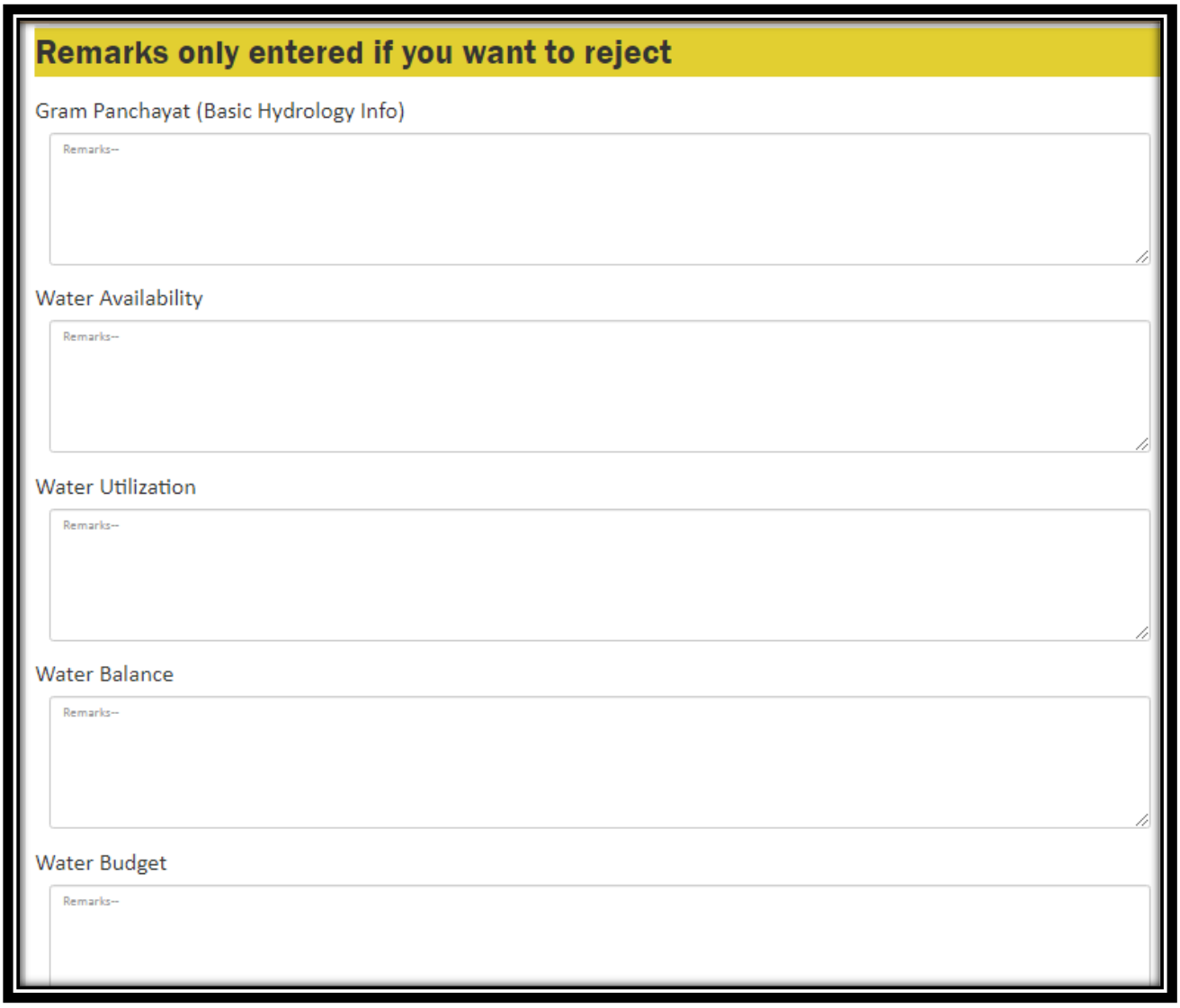
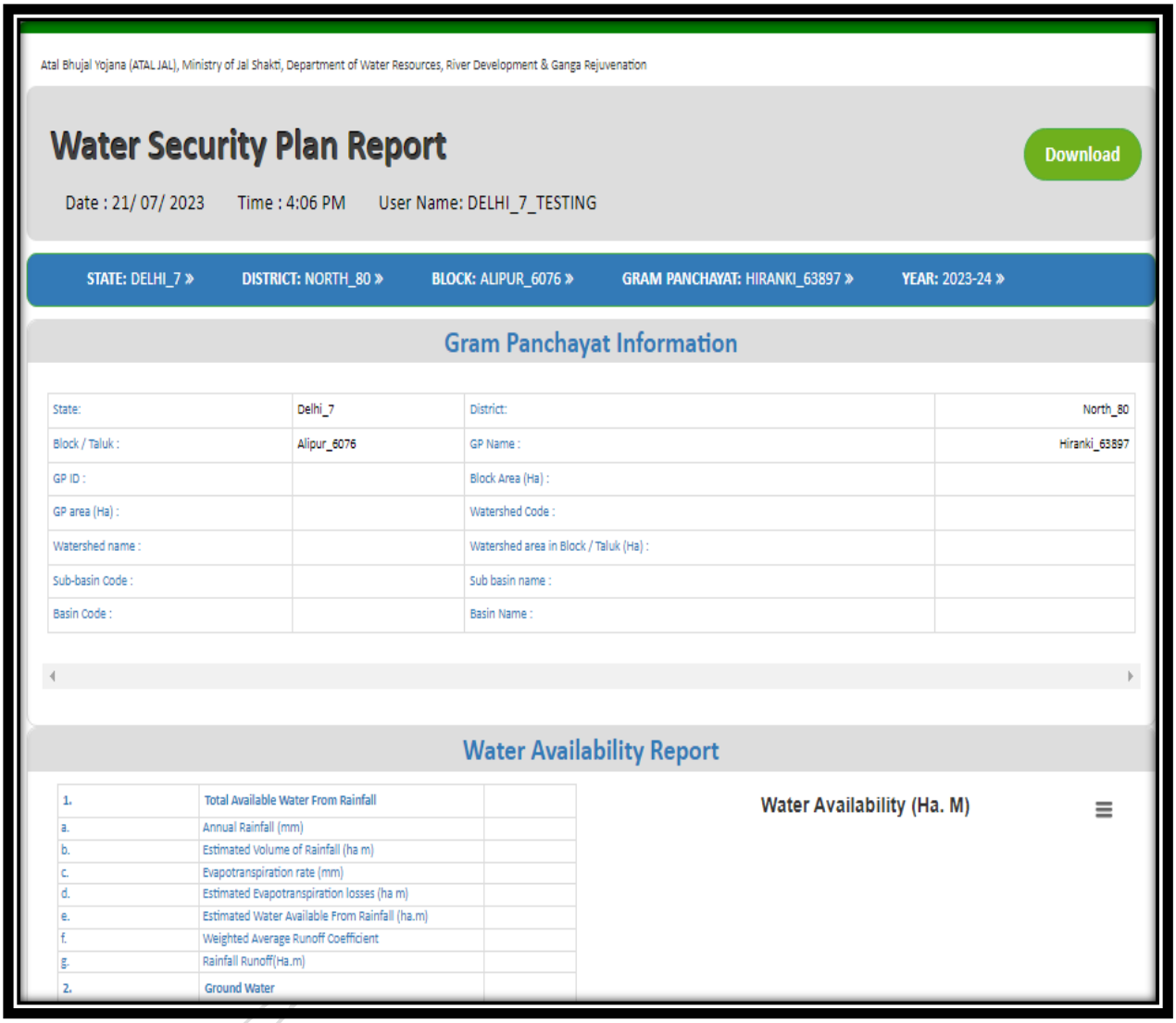


Fig: 9

10. FINAL WATER SECURITY PLAN REPORT

After the user completes data entry in the all WSP Sub modules, they can download the final report from this source.



Atal Bhujal Yojana (ATAL JAL), Ministry of Jal Shakti, Department of Water Resources, River Development & Ganga Rejuvenation

Water Security Plan Report

Download

Date : 21/ 07/ 2023 Time : 4:06 PM User Name: DELHI_7_TESTING

STATE: DELHI_7 » DISTRICT: NORTH_80 » BLOCK: ALIPUR_6076 » GRAM PANCHAYAT: HIRANKI_63897 » YEAR: 2023-24 »

Gram Panchayat Information

State:	Delhi_7	District:	North_80
Block / Taluk :	Alipur_6076	GP Name :	Hiranki_63897
GP ID :		Block Area (Ha) :	
GP area (Ha) :		Watershed Code :	
Watershed name :		Watershed area in Block / Taluk (Ha) :	
Sub-basin Code :		Sub basin name :	
Basin Code :		Basin Name :	

Water Availability Report

1.	Total Available Water From Rainfall	
a.	Annual Rainfall (mm)	
b.	Estimated Volume of Rainfall (ha m)	
c.	Evapotranspiration rate (mm)	
d.	Estimated Evapotranspiration losses (ha m)	
e.	Estimated Water Available From Rainfall (ha.m)	
f.	Weighted Average Runoff Coefficient	
g.	Rainfall Runoff(Ha.m)	
2.	Ground Water	

Water Availability (Ha. M)

DRAFT